UNIVERSITY OF BIRMINGHAM

School of Physics and Astronomy

DEGREE OF B.Sc. & M.Sci. WITH HONOURS

THIRD-YEAR FINAL EXAMINATION

03 00971

LH RADIATION AND RELATIVITY

SEMESTER 2 EXAMINATIONS 2021/22

Time Allowed: 1 hour 30 minutes

Answer ***two*** questions. If you answer more than two questions, credit will only be given for the best two answers.

The approximate allocation of marks to each part of a question is shown in brackets **[ ]**.

All symbols have their usual meanings.

A list of useful formulae is provided at the end of the paper.

Calculators may be used in this examination but must not be used to store text. Calculators with the ability to store text should have their memories deleted prior to the start of the examination.

A formula sheet and a table of physical constants and units that may be required will be found at the end of this question paper.

Answer **two** *questions. If you answer more than two questions, credit will only be given for the best two answers.*

1. The 4-velocity, u^μ, and the 4-acceleration, α^μ, of a particle are defined by

$$u^\mu = \frac{\mathrm{d}x^\mu}{\mathrm{d}\tau}, \qquad \alpha^\mu = \frac{\mathrm{d}u^\mu}{\mathrm{d}\tau} = \frac{\mathrm{d}^2x^\mu}{\mathrm{d}\tau^2}.$$

In a frame Σ with coordinate time t, the particle's 3-velocity is $\boldsymbol{v} = \mathrm{d}\boldsymbol{r}/\mathrm{d}t$ and its 3-acceleration is $\boldsymbol{a} = \mathrm{d}\boldsymbol{v}/\mathrm{d}t$.

(a) Explain what is meant by *coordinate time*, t, and *proper time*, τ, of the particle, and prove that $\mathrm{d}t = \gamma(v)\mathrm{d}\tau$, where $v = |\boldsymbol{v}|$, and $\gamma(v)$ is the Lorentz factor. **[4]**

(b) Show that u^μ and α^μ can be written in terms of $\boldsymbol{v}$ and $\boldsymbol{a}$, as **[6]**

$$u^\mu = \gamma(v)\,(c, \boldsymbol{v}), \qquad \alpha^\mu = \gamma^2(v)\,(0, \boldsymbol{a}) + \gamma^4(v)\frac{[\boldsymbol{v}\cdot\boldsymbol{a}]}{c^2}\,(c, \boldsymbol{v}).$$

(c) If a particle is moving with a constant angular velocity Ω about the z-axis, show that its 4-acceleration is **[3]**

$$\alpha^\mu = (0, \gamma^2(v)\boldsymbol{a}).$$

A relativistic electron, mass m, charge e, is in circular motion. It starts its motion in a magnetic field $\boldsymbol{B} = B\boldsymbol{e}_z$ with an initial velocity $\boldsymbol{v}_0 = v_0\boldsymbol{e}_x$. You can assume it moves in a circular orbit in the xy-plane with angular speed $\Omega = \Omega_0/\gamma(v)$, where $\Omega_0 = eB/mc$ is the cyclotron frequency. The Lorentz covariant expression for the rate of radiation loss of 4-momentum P^μ is

$$\frac{\mathrm{d}P^\mu}{\mathrm{d}\tau} = -\frac{2e^2}{3c^5}a^\nu a_\nu u^\mu.$$

(d) Find the total electron energy loss due to radiation, $\Delta\mathcal{E}$, during one revolution. Show that the ratio of $\Delta\mathcal{E}$ to the electron's kinetic energy $\mathcal{E}_\mathrm{k}$, is **[9]**

$$\left|\frac{\Delta\mathcal{E}}{\mathcal{E}}\right| = \frac{8\pi}{3}\Omega_0\frac{r_\mathrm{c}}{c}\frac{v_0^2\gamma^3}{2c^2(\gamma-1)}, \quad \text{where} \quad r_\mathrm{c} = \frac{e^2}{mc^2}.$$

(e) Find the nonrelativistic ($\gamma - 1 \ll 1$) and ultrarelativistic $\gamma \gg 1$) limits of the above expression, given that the maximum cyclotron frequency currently possible in the laboratory is $\Omega_0 \approx 10^{13}\mathrm{s}^{-1}$, and the classical electron radius $r_\mathrm{c} \approx 10^{-15}\mathrm{m}$. Under what conditions is the assumption valid that the electron orbit is circular? **[8]**

2. In this question work in units where the following constants are unity: $c = \mu_0 = \varepsilon_0 = 1$. Let $F^{\mu\nu}$ be the contravariant 4-tensor of the electromagnetic field, and $G_{\mu\nu}$ the dual field tensor.

   (a) Explain briefly why quantities such as $\eta_{\mu\nu}F^{\mu\nu}$, $F^{\mu\nu}F_{\mu\nu}$ and $F^{\mu\nu}G_{\mu\nu}$ are Lorentz invariant. **[3]**

   (b) Using the relations between components of $F^{\mu\nu}$ and $G^{\mu\nu}$ and the 3-vectors $\boldsymbol{E}$ and $\boldsymbol{B}$ of the electromagnetic field, or otherwise, express the three invariants in (a) in terms of $\boldsymbol{E}$ and $\boldsymbol{B}$. **[9]**

   (c) What are the values of these invariants for the field of an electromagnetic wave in free space? **[3]**

   The electromagnetic energy density W and the Poynting vector $\boldsymbol{S}$ can be expressed in terms of $\boldsymbol{E}$ and $\boldsymbol{B}$ as

$$W = \tfrac{1}{2}(E^2 + B^2), \qquad \boldsymbol{S} = \boldsymbol{E} \times \boldsymbol{B}.$$

   (d) What is the physical significance of $\boldsymbol{S}$? Using the result of (b) or otherwise, prove that the quantity $W^2 - |\boldsymbol{S}|^2$ is Lorentz invariant. **[8]**

   (e) In a certain inertial frame Σ the fields $\boldsymbol{E}$ and $\boldsymbol{B}$ make an acute angle θ and have equal magnitudes. For an observer in another inertial frame $\overline{\Sigma}$ the two fields are parallel. Find the angle θ, if $W/\overline{W} = 2$ are the electromagnetic energy densities in Σ and $\overline{\Sigma}$. **[7]**

3. The electromagnetic field of a particle of charge q, moving at a constant speed v in the laboratory frame Σ can be found by Lorentz transforming either the 4-potential or the field in its rest frame Σ' to Σ, choosing the frames to be in standard configuration.

   In the comoving frame Σ', the only non-zero component of the four-potential $A^{\mu'}$ and the electromagnetic field tensor $F^{\mu'\nu'} = \partial^{\mu'} A^{\nu'} - \partial^{\nu'} A^{\mu'}$ are given by

$$A^{0'} = \frac{\mu_0}{4\pi} \frac{q}{\sqrt{(x')^2 + (y')^2 + (z')^2}}, \qquad F^{i'0'} \equiv E^{i'}/c = \frac{\mu_0}{4\pi} \frac{q x^{i'}}{(r')^3}.$$

   (a) Find all nonzero components of $A^\mu = (V/c, \boldsymbol{A})$, expressing distances in terms of the coordinates in the frame Σ. **[9]**
   (Hint: recall the Lorentz transformations for $\Sigma' \to \Sigma$ and $\Sigma \to \Sigma'$ are represented by mutually inverse Lorentz transformation matrices.)

   (b) Hence show that the magnetic field $\boldsymbol{B}$ in Σ is given by **[6]**

$$\boldsymbol{B} = \frac{\mu_0}{4\pi} \frac{q}{c\gamma^2} \frac{\boldsymbol{v} \times \boldsymbol{r}}{[(x - vt)^2 + \gamma^{-2}(y^2 + z^2)]^{3/2}}.$$

   (c) Show that the same result can be obtained by substituting the Lorentz transformation formulas into the electromagnetic field tensor $F^{\mu'\nu'}$ thus finding the relevant components of $F^{\mu\nu}$ in Σ. **[9]**

   (d) The result in (b) is expressed in terms of the distance $\boldsymbol{r}$ between the observation point (where the field is measured) and the origin O of Σ. Rewrite $\boldsymbol{B}$ as a function of $\boldsymbol{R}$ and θ where $\boldsymbol{R} = \boldsymbol{r} - \boldsymbol{v}t$ is the distance between the moving particle and the observation point and θ is the angle between $\boldsymbol{R}$ and the direction of motion. **[6]**

Formula Sheet

- **Maxwell's equations** in 3-form, for electric field $\boldsymbol{E}$, magnetic field $\boldsymbol{B}$, electric current density $\boldsymbol{J}$ and charge density ρ, in empty space, with electric constant ε_0 and magnetic constant μ_0, where $c = 1/\sqrt{\varepsilon_0\mu_0}$,

$$\begin{aligned} \nabla\cdot\boldsymbol{B} &= 0, & \nabla\times\boldsymbol{E}+\dot{\boldsymbol{B}} &= 0, \\ \varepsilon_0\nabla\cdot\boldsymbol{E} &= \rho, & \mu_0^{-1}\nabla\times\boldsymbol{B}-\varepsilon_0\dot{\boldsymbol{E}} &= \boldsymbol{J}, \end{aligned}$$

- **Lorentz force equations** in in 3-form, for particle charge q, speed $\boldsymbol{v}$, momentum $\boldsymbol{p}$, and energy $\mathcal{E}$,

$$\frac{\mathrm{d}\boldsymbol{p}}{\mathrm{d}t} = q(\boldsymbol{E}+\boldsymbol{v}\times\boldsymbol{B}), \qquad \frac{\mathrm{d}\mathcal{E}}{\mathrm{d}t} = q\boldsymbol{E}\cdot\boldsymbol{v}.$$

- **Lorentz transformed EM fields** after a Lorentz boost by velocity $\boldsymbol{v}$,

$$\begin{aligned} \overline{\boldsymbol{E}} &= \gamma(\boldsymbol{E}+\boldsymbol{v}\times\boldsymbol{B}) - \frac{\gamma^2}{c^2(\gamma+1)}\boldsymbol{v}(\boldsymbol{v}\cdot\boldsymbol{E}), \\ \overline{\boldsymbol{B}} &= \gamma(\boldsymbol{B}-c^{-2}\boldsymbol{v}\times\boldsymbol{E}) - \frac{\gamma^2}{c^2(\gamma+1)}\boldsymbol{v}(\boldsymbol{v}\cdot\boldsymbol{B}) \end{aligned}$$

- **Electromagnetic potentials** with V the scalar potential, $\boldsymbol{A}$ the vector potential,

$$\boldsymbol{B} = \nabla\times\boldsymbol{A}, \qquad \boldsymbol{E} = -\dot{\boldsymbol{A}} - \nabla V.$$

- **Minkowski Metric**, given by $\eta_{\mu\nu}$ or $\eta^{\mu\nu}$, where all components zero except

$$\eta_{00} = \eta^{00} = +1, \qquad \eta_{jj} = \eta^{jj} = -1, \qquad \text{for } j = 1,2,3.$$

- **Spacetime position** x^μ transforms as contravariant 4-vector,

$$x^\mu = (x^0, x^1, x^2, x^3) = (ct, \boldsymbol{r}); \qquad x^\mu = (x_0, x_1, x_2, x_3) = (ct, -\boldsymbol{r}).$$

- **Covariant Derivative**, ∂_μ transforms as covariant 4-vector,

$$\partial_\mu = (\partial_0, \partial_1, \partial_2, \partial_3) = (c^{-1}\partial_t, \boldsymbol{\nabla}), \qquad \partial^\mu = (\partial^0, \partial^1, \partial^2, \partial^3) = (c^{-1}\partial_t, -\boldsymbol{\nabla}).$$

- **Lorentz transformation**, $\Lambda^{\overline{\mu}}_{\nu}$ from Σ to $\overline{\Sigma}$, for a boost by velocity $v = \beta c$ in standard configuration, has nonzero components

$$\Lambda^{\overline{0}}_{0} = \Lambda^{\overline{1}}_{1} = \gamma, \quad \Lambda^{\overline{0}}_{1} = \Lambda^{\overline{1}}_{0} = -\gamma\beta, \quad \Lambda^{\overline{2}}_{2} = \Lambda^{\overline{3}}_{3} = 1,$$

with the Lorentz factor $\gamma = \gamma(v) = (1-\beta^2)^{-1/2}$.

- **d'Alembertian operator**, a Lorentz scalar operator,

$$\Box^2 \;\equiv\; \eta^{\mu\nu}\partial_\mu\partial_\nu \;=\; c^{-2}\partial_t^2 - \nabla^2.$$

- **Faraday tensor** in matrix form

$$F_{\mu\nu} \;=\; \begin{pmatrix} 0 & E_x/c & E_y/c & E_z/c \\ -E_x/c & 0 & -B_z & B_y \\ -E_y/c & B_z & 0 & -B_x \\ -E_z/c & -B_y & B_x & 0 \end{pmatrix},$$

$$F^{\mu\nu} \;=\; \begin{pmatrix} 0 & -E_x/c & -E_y/c & -E_z/c \\ E_x/c & 0 & -B_z & B_y \\ E_y/c & B_z & 0 & -B_x \\ E_z/c & -B_y & B_x & 0 \end{pmatrix},$$

so $F_{0i} = E_i/c$ and $F^{ij} = -\varepsilon^{ijk}B_k$, for $i, j, k = 1, 2, 3$.
It follows from the 4-potential $A^\mu \;=\; (V/c, \boldsymbol{A})$ that

$$F_{\mu\nu} \;=\; \partial_\mu A_\nu - \partial_\nu A_\mu.$$

- **dual Faraday tensor** $G_{\mu\nu}$ is given by

$$G_{\mu\nu} \;=\; \tfrac{1}{2}\varepsilon_{\mu\nu\alpha\beta}F^{\alpha\beta}, \qquad G^{\mu\nu} \;=\; -\tfrac{1}{2}\varepsilon^{\mu\nu\alpha\beta}F_{\alpha\beta}.$$

Its components satisfy $G_{0i} = -B_i$, $G^{ij} = -\varepsilon^{ijk}E_k$ for $i, j, k = 1, 2, 3$.

- **Levi-Civita symbol** in 3D satisfies

$$\varepsilon_{ijk} \;=\; \varepsilon^{ijk} \;=\; \begin{cases} +1 & \text{if } \; ijk \text{ is an even permutation} \\ -1 & \text{if } \; ijk \text{ is an odd permutation.} \end{cases}$$

Here, even permutations consist of $123, 231, 312$, and odd permutations are $132, 213, 321$. It also satifies $\varepsilon_{ijk}\varepsilon^{imn} = \delta_j^m\delta_k^n - \delta_k^m\delta_j^n$, and

$$\boldsymbol{a}\cdot(\boldsymbol{b}\times\boldsymbol{c}) \;=\; \varepsilon^{ijk}a_ib_jc_k.$$

- **Levi-Civita symbol** in 4D satisfies

$$\varepsilon_{\mu\nu\rho\sigma} \;=\; -\varepsilon^{\mu\nu\rho\sigma} \;=\; \begin{cases} +1 & \text{if } \; \mu\nu\rho\sigma \text{ is an even permutation of } 0123 \\ -1 & \text{if } \; \mu\nu\rho\sigma \text{ is an odd permutation of } 0123 \\ 0 & \text{if } \; \text{anything else.} \end{cases}$$

The signs of the permutations of $0, 1, 2, 3$ are given by

$+$	0123,	0231,	0312,	1032,	1203,	1320,	2013,	2130,	2301,	3021,	3102,	3210,
$-$	0132,	0213,	0321,	1023,	1230,	1302,	2031,	2103,	2310,	3012,	3120,	3201.

Physical Constants and Units

Acceleration due to gravity	g	$9.81\,\mathrm{m\,s^{-2}}$
Gravitational constant	G	$6.674 \times 10^{-11}\,\mathrm{N\,m^2\,kg^{-2}}$
Ice point	T_{ice}	$273.15\,\mathrm{K}$
Avogadro constant	N_A	$6.022 \times 10^{23}\,\mathrm{mol^{-1}}$
		[*N.B.* 1 mole $\equiv 1$ *gram-molecule*]
Gas constant	R	$8.314\,\mathrm{J\,K^{-1}\,mol^{-1}}$
Boltzmann constant	k, k_B	$1.381 \times 10^{-23}\,\mathrm{J\,K^{-1}} \equiv 8.62 \times 10^{-5}\,\mathrm{eV\,K^{-1}}$
Stefan constant	σ	$5.670 \times 10^{-8}\,\mathrm{W\,m^{-2}\,K^{-4}}$
Rydberg constant	R_∞	$1.097 \times 10^{7}\,\mathrm{m^{-1}}$
	$R_\infty hc$	$13.606\,\mathrm{eV}$
Planck constant	h	$6.626 \times 10^{-34}\,\mathrm{J\,s} \equiv 4.136 \times 10^{-15}\,\mathrm{eV\,s}$
$h/2\pi$	$\hbar$	$1.055 \times 10^{-34}\,\mathrm{J\,s} \equiv 6.582 \times 10^{-16}\,\mathrm{eV\,s}$
Speed of light *in vacuo*	c	$2.998 \times 10^{8}\,\mathrm{m\,s^{-1}}$
	$\hbar c$	$197.3\,\mathrm{MeV\,fm}$
Charge of proton	e	$1.602 \times 10^{-19}\,\mathrm{C}$
Mass of electron	m_e	$9.109 \times 10^{-31}\,\mathrm{kg}$
Rest energy of electron		$0.511\,\mathrm{MeV}$
Mass of proton	m_p	$1.673 \times 10^{-27}\,\mathrm{kg}$
Rest energy of proton		$938.3\,\mathrm{MeV}$
One atomic mass unit	u	$1.66 \times 10^{-27}\,\mathrm{kg}$
Atomic mass unit energy equivalent		$931.5\,\mathrm{MeV}$
Electric constant	ϵ_0	$8.854 \times 10^{-12}\,\mathrm{F\,m^{-1}}$
Magnetic constant	μ_0	$4\pi \times 10^{-7}\,\mathrm{H\,m^{-1}}$
Bohr magneton	μ_B	$9.274 \times 10^{-24}\,\mathrm{A\,m^2}$ ($\mathrm{J\,T^{-1}}$)
Nuclear magneton	μ_N	$5.051 \times 10^{-27}\,\mathrm{A\,m^2}$ ($\mathrm{J\,T^{-1}}$)
Fine-structure constant	$\alpha = e^2/4\pi\epsilon_0\hbar c$	7.297×10^{-3}= 1/137.0
Compton wavelength of electron	$\lambda_c = h/m_e c$	$2.426 \times 10^{-12}\,\mathrm{m}$
Bohr radius	a_0	$5.2918 \times 10^{-11}\,\mathrm{m}$
angstrom	Å	$10^{-10}\,\mathrm{m}$
barn	b	$10^{-28}\,\mathrm{m^2}$
torr (mm Hg at $0\,^\circ$C)	torr	$133.32\,\mathrm{Pa}$ ($\mathrm{N\,m^{-2}}$)

Do not complete the attendance slip, fill in the front of the answer book or turn over the question paper until you are told to do so.

Important Reminders

- Coats/outwear should be placed in the designated area.
- Unauthorised materials (e.g. notes or Tippex) must be placed in the designated area.
- Check that you do not have any unauthorised materials with you (e.g. in your pockets, pencil case).
- Mobile phones and smart watches **must** be switched off and placed in the designated area or under your desk. They must not be left on your person or in your pockets.
- You are not permitted to use a mobile phone as a clock. If you have difficulty seeing a clock, please alert an Invigilator.
- You are not permitted to have writing on your hand, arm or other body part.
- Check that you do not have writing on your hand, arm or other body part – if you do, you must inform an Invigilator immediately
- Alert an Invigilator immediately if you find any unauthorised item upon you during the examination.

Any students found with non-permitted items upon their person during the examination, or who fail to comply with Examination rules may be subject to Student Conduct procedures.